Process Characterization Plan

A-Mab Drug Substance — Low-pH Viral Inactivation (Step 6) — PCP-006

Process Development / Manufacturing Science & Technology (MSAT)

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Approvals

Table 1: Approval record (synthetic; signatures not applied).

Role	Function	Name (synthetic)	Signature	Date
Author	Process Development / MSAT	—	—	—
Reviewer	Analytical Development	—	—	—
Reviewer	Manufacturing Science	—	—	—
Reviewer	Statistics	—	—	—
Approver	Quality Assurance	—	—	—

Abbreviations

AEX, anion exchange chromatography; ALCOA+, the data integrity attributes attributable, legible, contemporaneous, original and accurate, plus complete, consistent, enduring and available; AMV, analytical method validation; ANOVA, analysis of variance; BLA, Biologics License Application; CCD, central composite design; CEX, cation exchange chromatography; CPP, critical process parameter; CQA, critical quality attribute; DoE, design of experiments; DS, drug substance; HMW, high molecular weight; icIEF, imaged capillary isoelectric focusing; LIMS, laboratory information management system; LRF, log reduction factor; MSAT, Manufacturing Science and Technology; NOR, normal operating range; PAR, proven acceptable range; RSM, response surface methodology; SDM, scale-down model; SEC-HPLC, size exclusion high performance liquid chromatography; SOP, standard operating procedure; TCID50, fifty percent tissue culture infectious dose; XMuLV, xenotropic murine leukaemia virus.

1 Purpose and scope

Low-pH viral inactivation is the hold step that follows Protein A capture in the A-Mab purification train. The step brings the Protein A eluate to acid pH, holds it there for a defined time and then neutralises it to the condition at which cation exchange is loaded. Acid disrupts the lipid envelope of an enveloped virus and denatures the glycoproteins the virus uses to enter a cell, so a particle that has been held at the inactivation pH stays in the pool and can no longer infect. The step is one of three orthogonal steps that together carry the enveloped virus safety claim for the drug substance.

This plan describes the studies that will characterize the step for commercial manufacture at 15,000 L, and it fixes the acceptance and decision criteria before any of the data exist. The studies will be executed on a qualified scale-down model of the commercial hold. Their conclusions will be applied to the commercial step only to the extent that the qualification described in Section 5.1 supports, and that limit is stated again with each claim the report makes.

The scope of the plan is the acidification of the Protein A eluate, the hold itself and the neutralisation that ends it. 4 process parameters are in scope, of which 3 will be varied together in designed experiments and 1 will be assessed one factor at a time. Section 4.3 gives the basis for that split. 3 responses will be measured on every run, one for the viral safety claim and two for product quality, and they are listed with their acceptance criteria in Table 7.1.

Three things sit outside the scope of this plan. The step makes no clearance claim for host cell protein, DNA or leached Protein A, and the studies described here do not measure them. The virus clearance of the anion exchange and virus filtration steps is characterized in their own plans and reports, and the cumulative claim over the whole train is consolidated in PCMR-001. The commercial equipment qualification and the process performance qualification runs are covered by PTP-001 and are not repeated here.

2 Objectives

The studies described in this plan have five objectives.

(i) To qualify a scale-down model of the commercial hold, and to show that it reproduces the pH, the temperature and the hold time of the commercial step closely enough for its results to be applied to that step.

(ii) To measure the effect of inactivation pH, hold time and temperature on the log reduction of XMuLV, on the aggregate content of the neutralised pool and on the acidic charge variant content, over ranges deliberately wider than the normal operating ranges.

(iii) To fit a response-surface model to each of the 3 responses over the characterized region, and to establish the region within which every acceptance criterion in Table 7.1 is met at the same time.

(iv) To determine a proven acceptable range for each parameter against those criteria, by the two analyses set out in Section 8.

(v) To provide the parameter classification and the step contribution to the modular enveloped virus clearance claim that the control strategy needs, for consolidation in PCMR-001.

Objectives (i) to (iii) are met by the executed study. Objectives (iv) and (v) are met by the analysis of that study, and the rules for both are fixed in Section 7 and Section 8 so that they are not chosen after the results are seen.

3 Related documents

This plan is written under the master plan for the characterization campaign and against the risk assessment that scoped it. The documents it depends on, and the report that will carry its results, are listed in Table 3.1.

Table 3.1: Documents related to this plan.

Document ID	Title	Relationship
PTP-001	Process Transfer Plan (A-Mab Drug Substance)	Parent — transfer scope
RA-001	Pre-Characterization Process Risk Assessment (A-Mab Drug Substance)	Parent — risk basis for study scope
PCMP-001	Process Characterization Master Plan (A-Mab Drug Substance)	Parent — master plan
PCR-006	Process Characterization Report (Low-pH Viral Inactivation (Step 6))	Sibling — paired document
PCMR-001	Process Characterization Master Report (A-Mab Drug Substance)	Rolls up into

The procedures that govern the execution and the analysis of the studies are listed in Table 3.2. SOP-2009 governs the operation of the step at both scales, SOP-1001 governs the qualification of the scale-down model, SOP-1002 governs the design and the statistical analysis of the designed experiments, and SOP-4001 governs the classification of the parameters once the results are available. The analytical methods are listed separately in Section 5.3.

Table 3.2: Controlled procedures governing this study.

Reference	Title	Type
SOP-1001	Qualification of Scale-Down Models	SOP
SOP-1002	Design, Execution and Statistical Analysis of DoE Studies	SOP
SOP-2009	Operation of the Low-pH Viral Inactivation Step	SOP
SOP-4001	Process-Parameter Classification and Control-Strategy Definition	SOP
AMV-3010	N-Glycan Map (2-AB HILIC-UPLC)	Method validation

Reference	Title	Type
AMV-3011	Size-Variants (SEC-HPLC)	Method validation
AMV-3012	Host-Cell Protein ELISA	Method validation
AMV-3013	Charge Variants (icIEF)	Method validation
AMV-3014	Residual DNA (qPCR)	Method validation

4 Prior knowledge and quality risk basis

4.1 Unit-operation description and prior knowledge

The step receives the Protein A eluate, which is already at acid pH from the elution buffer. Acid is added to bring the pool to the inactivation pH, the pool is held at controlled temperature for the hold time, and base is then added to neutralise it for cation exchange. Inactivation of an enveloped virus under these conditions follows first-order kinetics. The rate constant rises as the pH falls and as the temperature rises, and the hold time integrates that rate, so the log reduction achieved by the step is set by the three parameters together.

The acid of the hold partially unfolds the antibody as well. At acid pH the molecule carries a large net positive charge and the second constant domain of its heavy chain loses its folded structure, which exposes hydrophobic surface. Molecules in that state associate, and the association continues for as long as they are held at low pH and proceeds faster when they are warmer or more concentrated. Aggregate formed during the hold is carried into cation exchange, which is the step that removes it. Time at low pH can also add acidic species through acid-catalysed reactions at labile residues, including succinimide formation and hydrolysis.

The low-pH hold is a platform step. It has been used for three previously licensed Novacyte antibodies (X-Mab, Y-Mab and Z-Mab) and its conditions have remained essentially unchanged across those products and through A-Mab development (CMC Biotech Working Group 2009). The mechanism of inactivation does not depend on the antibody, and the platform experience therefore transfers to A-Mab directly. Characterization is required nonetheless, because the log reduction credited to the step enters a cumulative viral safety claim and the claim must be supported by ranges that were themselves demonstrated.

Antibody precipitation has been observed at pH below the low end of the range to be studied, so the low pH edge of the study is placed where the pool remains soluble. The hold is operated near ambient temperature in a suite that does not control temperature tightly, so the temperature range studied is wider than the range the suite is expected to deliver.

4.2 Quality attributes in scope

The hold inactivates XMuLV, the enveloped model virus whose clearance is the one critical quality attribute this step delivers. It also raises the aggregate content of the pool and adds acidic charge variants to it, and both of those attributes are formed upstream. Table 4.1 gives all three with the drug substance acceptance criteria and the criticality assigned in RA-001. The Tool #1 column is the criticality score from the risk ranking tool used in RA-001, which multiplies the severity of the impact an attribute has on safety or efficacy by the uncertainty in that assessment, so a higher score is a higher criticality.

Table 4.1: Quality attributes in scope for this step, with drug substance acceptance and criticality.

CQA	Category	Acceptance	Criticality	Tool #1
Viral clearance — XMuLV (enveloped)	viral safety	16.7–30 log10 (cumulative)	VH	20
Aggregates (HMW)	size variant	0–5 % HMW (SEC)	H	60
Acidic charge variants (deamidation)	charge variant	18–40 % (CEX/icIEF)	VL	4

The XMuLV criterion in Table 4.1 is cumulative over the whole purification train. Low-pH inactivation, anion exchange and virus filtration each contribute an independent log reduction, and the modular claim is the sum of the three (International Council for Harmonisation 2023). The criterion that applies to this step alone is therefore not the cumulative figure but the contribution the step must deliver for the cumulative figure to hold, and Section 7 states how that contribution is derived.

Aggregate is a high criticality attribute that this step adds to rather than removes. The hold hands its aggregate content on to cation exchange, which is the polishing step that reduces it, so the criterion that applies here is an in-process limit rather than the drug substance specification of 5 %. Acidic charge variants are of very low criticality and are formed mainly in the production bioreactor. They are measured across the hold because acid-catalysed deamidation is mechanistically plausible over the times studied, and the study will show whether the addition is large enough to matter.

4.3 Risk-based prioritization of parameters

RA-001 ranked every parameter of this step for its potential impact on the attributes in Table 4.1 and for its potential to interact with the other parameters. The ranking placed inactivation pH, hold time and temperature in the group warranting multivariate evaluation, and A-Mab concentration in the group whose range can be supported by a univariate study. Table 6.1 gives the outcome as the study type of each parameter.

Inactivation pH was ranked highest because the inactivation rate depends on it most steeply of the three, and because the same acid conditions set how far the antibody is unfolded and therefore how fast aggregate forms during the hold. A change in pH therefore moves the inactivation rate of the virus and the rate at which the antibody associates in the same stroke, and the control range of the parameter is narrow because both rates change steeply across it. Hold time and temperature were ranked with it because a longer hold and a warmer hold each raise the log reduction of XMuLV and the aggregate content of the pool together. They were also ranked with it because hold time integrates a rate constant that pH and temperature both set, so a longer hold at a lower pH gives a larger log reduction than either change alone would predict from its main effect.

A-Mab concentration was ranked lower and was assigned to a univariate study. Association is a bimolecular event, so a more concentrated pool aggregates faster, but the concentration of the pool has no part in the inactivation chemistry. Concentration is also a property of the Protein A eluate rather than a parameter the operator sets, so its range is set by the capture step and is wide. A univariate scan across that range is sufficient to bound its effect on the two product quality responses.

5 Materials and methods

5.1 Scale-down model and its qualification

The studies will be run in bench-scale hold vessels operated under the same principles as the commercial hold. The model will match the commercial step in the pH the pool is titrated to, the temperature it is held at and the time it is held for, because a lower pH and a warmer hold raise both the rate at which the viral envelope is disrupted and the rate at which the antibody associates, and the hold time integrates both rates. It will also match the commercial step in the titration scheme, the mixing regime during acid and base addition and the material of the wetted surfaces. Volume is the scale-dependent quantity and is not matched. The ratio of surface to volume differs between the vessels, and the qualification is what establishes that the difference does not change the responses.

Qualification will follow SOP-1001 and will be executed before the characterization runs begin. Replicate scale-down runs will be performed at the set-point conditions on a single Protein A pool, and their results will be compared with the results of commercial-scale runs on comparable material. The comparison will cover the log reduction of XMuLV, the aggregate content of the neutralised pool, the acidic charge variant content and the pH and temperature actually achieved during the hold. The model will be accepted as representative when no statistically significant difference is found between scales at $\alpha = 0.05$ for the three responses, and when the achieved pH and temperature at small scale lie within the tolerance the commercial step is controlled to.

Failure of any of those comparisons would stop the study. In that event the cause should be investigated and the model modified before the characterization runs are executed, and the modification and its requalification would be reported in PCR-006. Results from an unqualified model will not be used to support a range.

5.2 Operation

Each run will be executed on material from a single Protein A eluate pool, so that the input to every design point is the same. The pool will be characterized before the study for aggregate content, acidic charge variant content and A-Mab concentration, and those values will be reported as the input condition of the study.

A run begins with the addition of acid to the pool under mixing until the target inactivation pH is reached and stable. The vessel is then held at the target temperature for the target hold time, with pH and temperature recorded throughout. At the end of the hold, base is added under mixing to bring the pool to the cation exchange load pH, and the neutralised pool is filtered. The recorded pH and temperature traces will be reported with the results, because the parameter values that enter the model are the achieved values and not the targets.

The clearance runs will be spiked with XMuLV. The spike will be added to the pool after acidification and before the hold begins, at the volume ratio fixed in SOP-2009, which is small enough that it does not change the pH or the antibody concentration of the hold. A sample of the spiked pool will be taken at the start of the hold to establish the input titre. Runs will be executed in randomized order within each design, and the operator, the pool lot and the analytical sequence will be recorded for each run.

5.3 Analytical methods

The three responses will be measured by the validated methods in Table 5.1. XMuLV infectivity will be determined by TCID50 under AMV-3017, aggregate content by SEC-HPLC under AMV-3011 and acidic charge variants by icIEF under AMV-3013.

Table 5.1: Validated analytical methods for the study responses.

Reference	Title	Type
SOP-1001	Qualification of Scale-Down Models	SOP
SOP-1002	Design, Execution and Statistical Analysis of DoE Studies	SOP
SOP-2003	Operation of the Production Bioreactor (Fed-Batch)	SOP
SOP-2004	Cell-Culture Media and Feed Preparation	SOP
SOP-3010	Released N-Glycan Mapping by 2-AB HILIC-UPLC	SOP
SOP-3011	Size-Variant Analysis by SEC-HPLC	SOP
SOP-3012	Host-Cell-Protein Quantitation by ELISA	SOP
SOP-3013	Charge-Variant Analysis by icIEF	SOP
SOP-3014	Residual Host-Cell DNA by qPCR	SOP
SOP-4001	Process-Parameter Classification and Control-Strategy Definition	SOP
AMV-3017	XMuLV Infectivity Titre (TCID50)	Method validation
AMV-3011	Size-Variants (SEC-HPLC)	Method validation
AMV-3013	Charge Variants (icIEF)	Method validation

The log reduction factor for XMuLV will be calculated as the difference between the log10 titre of the spiked pool at the start of the hold and the log10 titre of the sample taken at the end of the hold, each corrected for the volume of the fraction assayed. Where the end of hold sample gives no detectable infectivity, the titre will be reported as the limit of detection of the assay and the resulting log reduction will be reported as a greater-than value. A run reported that way constrains the model from below only, and the report will say so where it happens.

Aggregate and acidic charge variant samples will be drawn after neutralisation, because those are the values the cation exchange step receives. Samples will be assayed within the stability window established for the method, and every analytical sequence will carry a system suitability control and a pool control drawn from the study input material. The pool control is what allows drift between analytical sequences to be separated from an effect of the process parameters.

5.4 Sampling plan

The sampling plan is the same for every design point and is given in Table 5.2. Three samples are taken from each spiked run and two from each unspiked run.

Table 5.2: Samples taken from each characterization run.

Sample	Point in the run	Assay	Method
Input pool	Before acidification	Aggregate, acidic variants, concentration	AMV-3011, AMV-3013
Spiked load	After acidification, before the hold	XMuLV infectivity	AMV-3017
End of hold	At the end of the hold	XMuLV infectivity	AMV-3017
Neutralised pool	After neutralisation and filtration	Aggregate, acidic variants	AMV-3011, AMV-3013

Centre-point runs will be sampled identically to the factorial runs. Their replicate results are the estimate of pure error that the lack-of-fit test in Section 5.5 depends on, so no centre-point run may be excluded from the analysis for any reason other than a documented execution deviation.

5.5 Statistical methods

The designs will be analysed on coded factors, where the low edge of the characterization range is −1, the high edge is +1 and the design centre is 0. Coding puts the estimated effects on a common scale, so that an effect measures the change in the response over half the range studied and effects on different parameters can be compared directly.

The screening design will be analysed by a model containing the main effects and all two-factor interactions, which is 7 terms against 11 runs and leaves 4 residual degrees of freedom. Effects will be tested at $\alpha = 0.05$ and reported with their p-values.

The curvature of each response will be tested by comparing the mean of the centre points with the mean of the factorial points. Curvature that is significant at $\alpha = 0.05$ indicates that a model with quadratic terms is needed, which the response-surface design in Section 6.3 provides.

The response-surface design will be analysed by a full quadratic model containing the main effects, the two-factor interactions and the pure quadratic terms, which is 10 terms against 18 runs and leaves 8 residual degrees of freedom. The 4 centre-point replicates supply 3 degrees of freedom of pure error, and the remaining 5 degrees of freedom carry the lack of fit. The partition will be reported as an analysis of variance for each response, with the regression, residual, lack-of-fit and pure-error terms and their F ratios.

Model adequacy will be judged on four things, and all four will be reported for every response. These are the coefficient of determination and its adjusted form, the significance of the regression at $\alpha = 0.05$, the lack-of-fit F test against pure error, and the residual diagnostics. The diagnostics are the residuals plotted against the predicted values, a normal quantile plot of the residuals and the actual values plotted against the predicted values.

The screening model identifies which effects are present. The response-surface model is the predictive model, and it is the only model from which an operating region, a proven acceptable range or a prediction at an untested setting will be derived. The screening fit of a design this size is close to saturated, and nothing in this plan or in PCR-006 will claim predictive value for it.

The response-surface model predicts the mean level of each response. Where the predicted mean sits exactly on a criterion, about half of the material produced at that setting would be expected to fall on the wrong side of the criterion, so an operating region drawn on the predicted means carries no margin at its own edge. The Monte-Carlo analysis in Section 8 is what puts that margin back, and the report will not claim assurance from a predicted mean alone. All analyses will be run from the seeded analysis scripts under SOP-1002, so that the same data return the same result on re-execution.

6 Study design

6.1 Factors, ranges and study type

Table 6.1 gives the parameters in scope with their set-points, the ranges to be studied, the normal operating ranges and the study type assigned in Section 4.3. The plan states no classification for any parameter. Classification is an outcome of the study and will be assigned in PCR-006 under SOP-4001.

Table 6.1: Parameters, ranges to be studied, normal operating ranges and study type.

Parameter	Unit	Set-point	Range studied	NOR	Study type
Inactivation pH	pH	3.5	3.2–4	3.4–3.6	multivariate
Hold time	min	90	60–180	60–120	multivariate
Temperature	°C	21	15–25	19–23	multivariate
A-Mab concentration	g/L	20	5–35	5–35	univariate

The ranges studied are wider than the normal operating ranges in every case, and each edge is placed where it is for a reason of its own. Inactivation pH will be studied from 3.2 to 4 against a normal operating range of 3.4 to 3.6. The low edge is placed above the pH at which antibody precipitation has been observed on the platform, and the high edge is placed above the pH at which the platform expects inactivation to become slow within the shortest hold studied. Hold time will be studied from 60 to 180 min against a normal operating range of 60 to 120 min. The minimum is what a clearance claim must be assured at and the maximum is a deliberate challenge to the aggregate response, since it is a hold half again as long as the longest hold the process allows. Temperature will be studied from 15 to 25 °C, which brackets what a manufacturing suite delivers around a normal operating range of 19 to 23 °C.

The three multivariate factors are coded A to C for the design matrices, in the order given in Table 6.2. The same letters are used in the screening design, in the response-surface design and in the model terms reported in PCR-006.

Table 6.2: Coded factor legend for the designed experiments.

Code	Factor	Unit	Studied in
A	Inactivation pH	pH	screening + RSM
B	Hold time	min	screening + RSM
C	Temperature	°C	screening + RSM

6.2 Screening design

The screening design is a full factorial in the 3 multivariate factors with 3 centre-point replicates, which is 8 factorial runs and 11 runs in total. The planned coded matrix is given in Appendix A. A full factorial was chosen over a fractional design because three factors is small enough that the full design costs only 8 runs, and because a full factorial confounds no main effect with an interaction. Hold time integrates a rate constant that pH sets, so pH and hold time interact in the log reduction, and that interaction must be estimated cleanly and not aliased with a main effect.

The centre points serve two purposes. Their replicate results give the pure error of the step and the analytical methods together, which is the reproducibility figure the report will quote. Their mean, compared with the mean of the factorial points, tests each response for curvature. Curvature is expected in the XMuLV response, because first-order inactivation kinetics are linear in the log reduction against time but not against pH.

The screening study will run first and its outcome decides nothing about which factors continue. All 3 factors will be carried into the response-surface design whatever the screening result. A lower pH, a longer hold and a warmer hold each raise the log reduction of XMuLV and the aggregate content of the pool, and the control range of each has to be justified in the report whether or not the screening design resolves its effect.

6.3 Response-surface design

The response-surface design is a face-centred central composite design in the same 3 factors. It comprises 8 factorial runs, 6 axial runs and 4 centre-point replicates, which is 18 runs in total. The planned coded matrix is given in Appendix B. Placing the axial points on the faces of the cube, at coded ±1 rather than beyond it, keeps every run inside the ranges in Table 6.1. The design therefore never requires a setting outside the range the plan commits to studying, and the fitted model is never asked to extrapolate.

The centre of the design is inactivation pH 3.6, hold time 120 min and temperature 20 °C, and the 4 centre-point replicates are run at those settings. The axial runs are what allow the pure quadratic terms to be estimated.

The two designs will be executed on the same Protein A pool where the material supply allows it. Where a second pool is required, the pool will be treated as a block in the analysis and the block effect will be reported. Executing both designs gives 29 runs on the same step. The report will present the two designs separately, because the screening design estimates effects and only the response-surface design predicts a response at a setting that was not run.

6.4 Univariate assessment

A-Mab concentration will be assessed one factor at a time. The pool will be adjusted to settings across the range 5 to 35 g/L, and the other three parameters will be held at their set-points. Aggregate content, acidic charge variant content and XMuLV log reduction will be measured on every run.

A univariate design cannot resolve an interaction between concentration and the other parameters, and it is not asked to. The mechanism gives concentration no part in the inactivation chemistry, and its effect on aggregation is expected to act independently of the pH and the temperature of the hold. Should the univariate results contradict that expectation, the report will say so and the parameter will be brought into a multivariate study rather than supported by this design.

7 Acceptance and decision criteria

The criteria in this section are fixed before the study is executed. They are the yardsticks against which the results in PCR-006 will be judged, and they will not be revised after the data are seen.

7.1 Acceptance criteria for the responses

Table 7.1 gives the criterion that applies to each response, the quality attribute it belongs to and the basis on which the criterion was derived.

Table 7.1: Acceptance criteria for the study responses and their basis.

Response	Quality attribute	Acceptance criterion	Criterion basis
Aggregates (HMW, %)	Aggregates (HMW)	≤ 2.17	in-process limit
XMuLV LRF ($\log_{10}$)	Viral clearance — XMuLV (enveloped)	≥ 4.93	back-calculated step contribution
Acidic variants	Acidic charge variants (deamidation)	≤ 40	drug substance specification

The XMuLV criterion is the contribution this step must make for the cumulative claim to hold. The drug substance requires a cumulative enveloped virus clearance of at least 16.7 log10 across the train. Anion exchange and virus filtration are credited with their own demonstrated contributions, and the difference between the cumulative requirement and that credit is what this step must deliver, which is 4.93 log10.

The aggregate criterion is an in-process limit of 2.17 %, not the drug substance specification of 5 %. The neutralised pool is an intermediate, and cation exchange reduces its aggregate content before the material becomes drug substance. The in-process limit is derived by carrying the drug substance specification back through the aggregate clearance the steps after the hold deliver, subtracting the aggregate those steps add, and then dividing by a safety factor so that the drug substance does not depend on every downstream step delivering its nominal clearance exactly. Acidic charge variants are judged against the drug substance specification of 40 %, because no downstream step of the train modifies them.

7.2 Model adequacy criteria

A response-surface model will be accepted as adequate for prediction when its regression is significant at $\alpha = 0.05$, its lack of fit is not significant at $\alpha = 0.05$ against pure error, and its residual diagnostics show no structure. A model that fails any of these will not be used to define an operating region or a proven acceptable range for its response.

Two outcomes are anticipated and will be handled as follows. Where a response varies so little across the region that its regression is not significant, the report will state that the parameters have no detectable effect on it over the ranges studied and will support the ranges from the observed values instead of from a model. Where the pure error of a response is very small relative to its fitted values, the lack-of-fit F ratio may become large for differences too small to matter, and the report should then judge adequacy on the size of the residuals against the acceptance criterion rather than on the F test alone. Neither outcome is a reason to refit a different model.

7.3 Decision criteria for the operating region

The operating region of the step is the part of the characterized region in which every criterion in Table 7.1 is met at the same time. It will be established from the fitted response-surface models by evaluating all 3 responses over a grid spanning the ranges in Table 6.1 and retaining the settings that satisfy every criterion. The proportion of the characterized region that satisfies every criterion will be reported.

The study will be considered to have met its objectives when the scale-down model is qualified, when both designs are executed as planned or with documented deviations whose impact is assessed, when an adequate model is obtained for each response or the absence of an effect is demonstrated, and when the normal operating ranges in Table 6.1 lie inside the operating region. If a normal operating range is found to extend outside the operating region, the report will define the edge of failure, and the normal operating range should be narrowed in the control strategy rather than the criterion relaxed.

8 Proven acceptable ranges (planned analysis)

A proven acceptable range is the range of a parameter over which the step has been demonstrated to meet its acceptance criteria (International Council for Harmonisation 2009). For each quality attribute in Table 4.1 and each multivariate parameter, a proven acceptable range will be derived from the fitted response-surface model of the corresponding response, against the criterion for that response in Table 7.1.

Two analyses will be run for every combination of attribute and parameter. The first holds the other parameters at their set-points and scans the parameter of interest across its characterization range. It answers what the parameter may do while the rest of the step is running as intended. The second propagates the normal operating ranges of the other parameters. At each of 81 points across the characterization range of the parameter of interest, 2,000 Monte-Carlo draws are taken with the other parameters sampled within their normal operating ranges, the fitted model is evaluated at each draw with its prediction uncertainty included, and the criterion is applied to the result. It answers what the parameter may do while the rest of the step varies as it is permitted to.

The proven acceptable range returned by each analysis is the contiguous interval around the design centre over which the criterion is met. An interval is reported only where it is contiguous, because a parameter that meets a criterion in two separated intervals is not supported by a range and should be reported as an edge of failure instead. Where the two analyses disagree, the narrower range is the one that will be carried into the control strategy, and both will be shown in the report.

Each proven acceptable range will be reported with a plot of the response against the parameter, with the acceptable region shaded green, the set-point marked and the normal operating range marked. The plot shows how steeply the response approaches its criterion. A parameter whose response crosses its criterion steeply needs tighter control than one whose response approaches it slowly, even where the two proven acceptable ranges are equally wide.

Two limits apply to every proven acceptable range this study will produce. The range is proven only within the characterization range that was studied, and a range reported as the full characterization range means that no edge was found inside it rather than that none exists. The range is also conditioned on the scale-down model, and it transfers to the commercial step only under the qualification in Section 5.1.

9 Data management and integrity

Raw data from the study will be recorded contemporaneously and will meet the ALCOA+ attributes throughout. Process data from the hold vessels, which are the pH and temperature traces and the recorded times, will be captured by the instrument systems and retained as the original electronic records. Analytical results will be reported from the validated methods into the laboratory information management system, and the instrument data files will be retained under the record retention policy of the site.

Manual entries will be made in the controlled study notebook and will carry the date and the identity of the person making them. Corrections will be made so that the original entry remains legible, with the reason recorded. No result may be excluded from the analysis without a documented investigation, and any exclusion will be reported in PCR-006 with its justification.

The statistical analysis will be executed from version-controlled scripts under SOP-1002 against a locked dataset. The dataset, the script version and the software environment will be recorded in the report, so that the analysis can be re-executed and returns the same result. A second reviewer independent of the analyst will verify the transcription of the results into the report and the correctness of every value quoted from the analysis.

10 Roles and responsibilities

The functions responsible for the study and their responsibilities are given in Table 10.1. The named signatories are recorded in Table 1.

Table 10.1: Functions and responsibilities for the study.

Function	Responsibility
Process Development / MSAT	Authors this plan, executes the scale-down runs, interprets the results and authors PCR-006
Analytical Development	Executes the size variant and charge variant methods and maintains their validated state
Virology	Executes the XMuLV spiking runs and the infectivity assay, and calculates the log reduction factors
Manufacturing Science	Confirms that the scale-down model represents the commercial step and that the ranges studied cover commercial operation
Statistics	Approves the designs, executes the analysis and approves the model adequacy and range conclusions
Quality Assurance	Approves this plan and PCR-006 and confirms that the study was executed under the governing procedures

Process Development owns the study and is accountable for its execution against this plan. Statistics is accountable for the analysis and for the conclusions drawn from the models, and no range or classification will be reported without that approval. Quality Assurance approves the plan before execution begins and approves the report before its conclusions enter the control strategy.

11 Deliverables and schedule

The deliverable of this plan is the Process Characterization Report for the step, PCR-006. It will report the qualification of the scale-down model, the executed designs, the fitted models with their adequacy, the operating region, the proven acceptable ranges, the classification of each parameter and the contribution of the step to the enveloped virus clearance claim. Any deviation from this plan will be recorded in that report with its impact on the conclusions.

The study runs in four phases in the order given in Table 11.1. No calendar dates are fixed in this plan. Each phase begins on completion of the one before it. The characterization runs will not begin until the scale-down model has been accepted, because results from an unqualified model cannot support a range.

Table 11.1: Study phases and the gate at the end of each.

Phase	Content	Gate to the next phase
Qualification	Scale-down model qualification	Model accepted under SOP-1001
Screening	Screening design and univariate assessment	Designs executed and data locked
Response surface	Response-surface design	Design executed and data locked
Analysis	Proven acceptable ranges, classification and reporting	PCR-006 approved

The results of this study feed the master report PCMR-001, which consolidates the cumulative viral clearance claim across the three orthogonal steps and the control strategy across the train. PCR-006 is therefore required before the cumulative claim can be closed.

12 Risks and assumptions

The study rests on four assumptions and carries four risks, and each is stated here so that the report can confirm or correct it.

The first assumption is that the scale-down model represents the commercial hold. It is tested directly by the qualification in Section 5.1, and the risk it carries is that a difference between scales is present but too small to detect in replicate runs. The consequence would be a range that is proven at small scale and narrower at commercial scale. Confirmation at commercial scale during process performance qualification is what bounds that risk, and this plan claims nothing about the edges of the ranges at commercial scale.

The second assumption is that a single Protein A pool represents the material the step will receive. The pool fixes the input aggregate content, the input acidic variant content and the antibody concentration, and a pool with an unusual input would move all the responses together. The risk is mitigated by characterizing the pool before the study and by reporting its values as the input condition, so that a reader can see what the model is conditioned on. The univariate assessment in Section 6.4 covers the concentration part of that dependence across the full range the capture step delivers.

The third assumption is that the spiked runs behave as the unspiked step does. The spike is a virus preparation added to the pool, and it carries its own matrix. The volume ratio in SOP-2009 keeps that matrix to a small fraction of the hold, and the product quality responses of the spiked runs will be compared with those of the qualification runs to confirm that the spike did not shift them.

The fourth assumption is that the analytical methods resolve the differences the study needs to see. The risk is largest for the infectivity assay near complete inactivation, where the end of hold titre may fall below the limit of detection and the log reduction can only be reported as a greater-than value. Several such runs at the low pH corner would flatten the fitted surface there and would make the model conservative rather than wrong. The report will identify any run reported that way and will state what it does to the model.

One further risk is procedural. Material supply may force the two designs onto different Protein A pools, which introduces a block effect. Section 6.3 sets out how that would be handled, and the analysis would report the block effect rather than absorb it into the parameter estimates.

13 Approvals

This plan becomes effective when it has been approved by the functions recorded in Table 1. Execution of the studies described here may not begin before that approval. Amendments to the plan will be issued as a new version and approved by the same functions, and any amendment made after execution has begun will be reported in PCR-006 with the reason for it.

14 Appendices

The appendices give the planned design matrices in coded form. No results appear in them, because none exist at the time this plan is issued. The natural-unit setting for each coded level follows from the range of the factor in Table 6.1, where coded -1 is the low edge of the range studied, coded $+1$ is the high edge and coded 0 is the midpoint between them. The factor letters are those of Table 6.2.

14.1 Appendix A — Planned screening design matrix

Table 14.1: Planned screening design matrix in coded units. Runs will be executed in randomized order.

Run	Type	A	B	C
1	factorial	-1	-1	-1
2	factorial	-1	-1	1
3	factorial	-1	1	-1
4	factorial	-1	1	1
5	factorial	1	-1	-1
6	factorial	1	-1	1
7	factorial	1	1	-1
8	factorial	1	1	1
9	center	0	0	0
10	center	0	0	0
11	center	0	0	0

14.2 Appendix B — Planned response-surface design matrix

Table 14.2: Planned response-surface design matrix in coded units. Runs will be executed in randomized order.

Run	Type	A	B	C
1	factorial	-1	-1	-1
2	factorial	-1	-1	1
3	factorial	-1	1	-1
4	factorial	-1	1	1

Run	Type	A	B	C
5	factorial	1	-1	-1
6	factorial	1	-1	1
7	factorial	1	1	-1
8	factorial	1	1	1
9	axial	-1	0	0
10	axial	1	0	0
11	axial	0	-1	0
12	axial	0	1	0
13	axial	0	0	-1
14	axial	0	0	1
15	center	0	0	0
16	center	0	0	0
17	center	0	0	0
18	center	0	0	0

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